

## Egypt Car Accidents Analysis (2022 – 2024)

Power BI • Excel • Python • Data Analysis • Forecasting

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### Project Description

This project analyzes car accidents in Egypt from **2022 to 2024** using real datasets collected from the official website of the **Central Agency for Public Mobilization and Statistics (CAPMAS)**.

The project focuses on analyzing:

- Accident trends
- Injuries and deaths
- Emergency statistics
- Road safety patterns

The project also includes forecasting models using Python to predict future accident trends and support data-driven decision-making.

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### Project Objective

- Analyze car accident statistics in Egypt between 2022 and 2024
  - Identify trends in injuries and deaths
  - Compare accident categories and patterns
  - Discover insights related to:
    - Gender
    - Age groups
    - Vehicle types
    - Road types
    - Monthly statistics
  - Build interactive dashboards for storytelling and visualization
  - Predict future accident trends using Machine Learning
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### Project Resources

| Resource            | Link                               |
|---------------------|------------------------------------|
| Google Drive Folder | <a href="#">Open Project Files</a> |

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## Tools & Technologies

| Tool            | Purpose                                                 |
|-----------------|---------------------------------------------------------|
| Excel           | Data storage and organization                           |
| Power BI        | Data cleaning, transformation, analysis, and dashboards |
| Python          | Forecasting & Machine Learning                          |
| Figma           | Dashboard UI/UX Design                                  |
| Adobe Photoshop | Editing icons and dashboard assets                      |
| PowerPoint      | Presentation & Storytelling                             |

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## Project Workflow

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### Data Collection

- Collected datasets from the official CAPMAS website
  - Gathered:
    - 3 PDF reports (2022, 2023, 2024)
    - Population datasets for the same years
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### Understanding the Data

- Studied all tables carefully
- Determined the important datasets required for analysis
- Selected datasets related to:

- Injuries
- Deaths
- Vehicle Types
- Road Types
- Gender
- Age Groups
- Monthly Statistics

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### **Data Cleaning & ETL Process**

The datasets were imported into **Power BI Power Query** and processed using ETL techniques:

- Extract
- Transform
- Load

### **Challenges During Cleaning**

- Headers existed inside rows
- Large amount of null values
- Inconsistent table structures
- Different spellings for governorate names

### **Cleaning Process**

- Cleaned each year separately
- Standardized governorate names and text formatting
- Fixed data types
- Removed unnecessary rows and columns
- Prepared clean datasets for analysis

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### **Data Merging & Structuring**

Initially, the project contained:

- 30 Tables
  - 15 Injuries Tables
  - 15 Deaths Tables

### **First Merge**

Merged the same category across all years.

#### **Example**

- Injuries by Gender
- Injuries by Vehicle Type
- Injuries by Road Type
- Monthly Statistics

### **Result**

30 Tables → 10 Tables

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### **Final Merge**

Merged injuries and deaths datasets for each category.

### **Final Output**

10 Tables → 5 Final Tables

The data then became fully ready for analysis and visualization.

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## **Dashboard Development**

To improve performance and project organization, the project was divided into two separate Power BI files.

### **Data Cleaning File**

Used for:

- ETL Process
- Cleaning
- Data preprocessing

### **Dashboard & Analysis File**

Used for:

- DAX Measures
- Dashboard Development
- Visualizations
- Analysis

This approach improved dashboard performance and workflow organization.

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### **Data Preparation & Analysis**

- Organized and prepared the datasets for analysis inside Power BI
  - Built relationships between tables to improve dashboard performance
  - Created DAX measures for:
    - KPIs
    - Comparisons
    - Percentages
    - Dynamic calculations
  - Prepared the final datasets for dashboard visualizations and reporting
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### **Dashboard Design**

#### **Design Tools**

- Figma → Dashboard Layout & UI/UX
  - Photoshop → Editing icons and visual assets
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### **Dashboard Pages**

| Page       | Description                 |
|------------|-----------------------------|
| Intro Page | Project introduction        |
| Overview   | General accident statistics |

| Page                      | Description                      |
|---------------------------|----------------------------------|
| Injuries Analysis         | Injury trends and analysis       |
| Deaths Analysis           | Death trends and insights        |
| Comparison Analysis       | Comparisons between categories   |
| Ambulance Analysis        | Emergency and ambulance analysis |
| Ambulance Points Analysis | Ambulance locations and coverage |

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### 🌟 Dashboard Features

- ✓ Interactive Filters & Slicers
- ✓ Tooltips for Important Charts
- ✓ Interactive Navigation
- ✓ KPI Cards
- ✓ Dynamic Comparisons

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### 🖼️ Dashboard Preview

Add dashboard screenshots here

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### 🤖 Forecasting & Machine Learning

After completing the dashboard analysis, the project moved to Python for forecasting.

### 🔬 Steps Performed

- Exported clean data from Power BI
- Tested model performance and accuracy
- Built forecasting models for:
  - Accident trends
  - Injury rates

- Death rates

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### Forecasting Results

The forecasting models indicated that:

- Accident rates are expected to increase over time
- Death rates are also expected to increase in future years

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### Key Insights

- Accident rates increased across the analyzed years
- Male injury rates were significantly higher
- Certain road types recorded higher accident frequencies
- Forecasting models predicted continued increases in accidents and deaths

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### Presentation & Storytelling

The final presentation was created using **PowerPoint** and focused on:

- Problem Explanation
- Key Insights
- Dashboard Storytelling
- Forecasting Results
- Recommendations

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### Team Members & Responsibilities

| Team Member                  | Role                                                                               |
|------------------------------|------------------------------------------------------------------------------------|
| Mohamed Nasser Abd El Basset | Team Leader, Presentation, Project Coordination, and Supporting All Project Phases |
| Ahmed Abdelrhman Hassan      | Forecasting & Machine Learning using Python                                        |

| Team Member                | Role                                              |
|----------------------------|---------------------------------------------------|
| Youssef Ahmed Ibrahim      | DAX Measures & Calculations                       |
| Sara Basher Seleem         | Data Cleaning, Data Preparation, and Data Storage |
| Mariam Mohammed Abdelmalek | Data Cleaning, Data Preparation, and Data Storage |
| Tasneem Reda Gomaa         | Dashboard Development & Dashboard Design          |

### Instructor

**ENG. Karim Bakly**

### Project Timeline

| Phase                                                                                                                    | Duration |
|--------------------------------------------------------------------------------------------------------------------------|----------|
|  Data Collection                      | 2 Days   |
|  Understanding the Data               | 1 Week   |
|  Data Cleaning, Preparation & Merging | 1 Month  |
|  DAX Measures & Calculations          | 1 Week   |
|  Dashboard Design & Development       | 3 Days   |
|  Forecasting & Machine Learning       | 2 Days   |
|  Presentation & Storytelling          | 3 Days   |

### Total Project Duration

**Approximately 2 Months**



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### **Final Outcome**

The project successfully transformed raw governmental accident reports into a complete analytical solution that includes:

-  Clean and structured datasets
-  Interactive Power BI dashboards
-  Forecasting & Machine Learning models
-  Professional storytelling presentation

The final solution provides valuable insights into road accidents in Egypt and supports future road safety improvements and data-driven decisions.

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### **Project Status**

 **Completed Successfully**